

Neureset Use Cases

Date and Time Option (UC1)

Primary Actor:

- User/Patient

Stakeholders and Interests:

- User/Patient
- Neureset Device (automatic program)

Pre-condition:

- The device is powered on

Post-condition:

- The user/patient is able to set the date and time

Main success scenario:

1. User turns on the Neureset device
2. To set the date and time, the user/patient selects the “Date and Time” option from main menu
3. The user/patient sets the date and time to accurately track the sessions

New Neurofeedback Therapy Session (UC2)

Primary Actor:

- User/Patient

Stakeholders and Interests:

- User/Patient
- Neureset Device (automatic program)

Pre-condition:

- The Neuroset device is already charged and ready for any usage
- An EEG headset is properly positioned on the user/patient

Post-condition:

- With the device correctly positioned, the Neurofeedback therapy session begins

Main success scenario:

1. The user/patient powers the Neureset device on
2. The user/patient then selects the “New Session” option from the device menu
3. A check for proper electrode contact is made, indicated by a blue light. If secure, the session timer and progress bar initiate

Extensions:

- 3a. If the device cannot establish proper contact initially
 - The user/patient needs to adjust the headset and attempt again
- 3b. If the contact is lost, a red light flashes and the device emits a beeping noise
 - The user/patient needs to adjust the headset and attempt again until contact is re-established
 - If no contact is re-established within 5 minutes, the device will turn off

Establish Baseline (UC3)

Primary Actor:

- User/Patient

Stakeholders and Interests:

- User/Patient
- Neureset Device (automatic program)

Pre-condition:

- Neurofeedback session has begun with the device correctly positioned

Post-condition:

- A baseline to calculate the average dominant frequency over approximately one minute is established at all of the 21 sites
- The treatment can now commence

Main success scenario:

1. The device automatically commences to establish a baseline average frequency for the first of the 21 sites, and this process is indicated with a green light
2. This process is sequentially repeated for each of the remaining 20 sites, in order to ensure a comprehensive baseline setup for the brain
3. Once completed, the device then compiles the baseline data, readying it for the treatment phase

Treatment Application (UC4)

Primary Actor:

- User/Patient

Stakeholders and Interests:

- User/Patient
- Neureset Device (automatic program)

Pre-condition:

- The baseline frequencies have been established
- Treatment commences

Post-condition:

- The treatment, at the first EEG site, is now applied in accordance with the established baselines and is completed within one second

Main success scenario:

1. With the baselines being established, the device can commence the treatment application by selecting the first EEG site
2. After the first site's treatment, this process is then repeated automatically to the next site
3. The user/patient is kept informed of the session time and progress via a timer showing the approximate time remaining, and a progress bar indicated by a percentage of treatment complete in the UI

Completion Of Neurofeedback Session (UC5)

Primary Actor:

- User/Patient

Stakeholders and Interests:

- User/Patient
- Neureset Device (automatic program)

Pre-condition:

- The treatment has been applied at the first EEG site

Post-condition:

- All of the 21 sites have received their respective treatment to address every region of the brain
- The session has been completed
- The data from the session is now available for further review/progress

Main success scenario:

1. All 21 sites have been treated
2. Once the treatment for all 21 sites is completed, the Neureset device signals the end of the session with an auditory and visual cue
3. The device then automatically saves the data of the session, and this includes the brainwave frequencies of each site
4. The user/patient is prompted to either review the session on the device or to proceed to the other options
5. The device then returns to its home menu, essentially ready for a new session to commence or to be powered off
6. The user then removes the EEG headset, putting an end to the neurofeedback therapy session

Session Log Option (UC6)

Primary Actor:

- User/Patient

Stakeholders and Interests:

- User/Patient
- Neureset Device (automatic program)

Pre-condition:

- Some treatment sessions have been applied to the user

Post-condition:

- The user/patient is able to scroll through the session log

Main success scenario:

1. User powers on the device
2. To view past sessions, the user/patient can select the “Session Log” option
3. The date and time of the previous sessions is displayed
4. The user/patient can scroll through the list of sessions for different date and time

Neureset Use Case Diagram

